CHAPTER

9

MEASURES OF DISPERSION

OBJECTIVES

After studying the material in this chapter, you should be able to :

- *Understand the meaning of the term dispersion.*
- *Know the requisites of an ideal measure of dispersion.*
- *Differentiate between relative and absolute measures of dispersion.*
- *Identify different types of measures of dispersion.*
- *Calculate various measures of dispersion, such as the range, quartile deviation, mean deviation, standard deviation and variance.*
- *Know the merits and demerits of different measures of dispersion.*
- *Appreciate properties of standard deviation.*
- *Understand the concept of coefficient of variation.*

9.1 INTRODUCTION

The various measures of central tendency discussed in the previous chapter do not by themselves give an adequate description of the data. Regardless of the measure that may be used, it does not tell us whether the data are grouped closely together or spread over a large range of values. It is quite possible to have two or more sets of observations with the same mean but still differ considerably in the variability of their measurements about the average. For example, consider the following three sets of observations, each containing 9 items :

										Total	Mean
Set A	:	10	10	10	10	10	10	10	10	10	90	10
Set B	:	6	7	8	9	10	11	12	13	14	90	10
Set C	:	2	4	6	8	10	12	14	16	18	90	10

All the three sets of data have the same mean but otherwise they are quite different. Each set has a sum of 90, but in the first set this sum is dispersed among all the values. In fact, none of the observations in set *A* deviates from the mean and we say that there is no dispersion at all in set *A*. However, comparing the observations in sets *B* and *C*, it is quite obvious that set *B* is more uniform than set *C*. We say that the variability or the dispersion of the observations from the average is less for set *B* than for set *C*. Thus we need some measure of dispersion of a set of numbers. The dispersion of a data measures the degree to which numerical data tend to spread about an average value.

The measures of central tendency are, therefore, inadequate to describe the data completely. They must be supported and supplemented by some other measures.

9.2 DISPERSION

In the following we shall give some important definitions of dispersion.

1. *Dispersion is the measure of the variations of the items.* — **A.L. Bowley**
2. *Dispersion is the measure of the extent to which the individual items vary.* —**L.R. Connor**
3. *The degree to which numerical data tend to spread about an average value is called the variation or dispersion of the data.* — **Spiegel**
4. *Dispersion or spread is the degree of the scatter or variation of the variables about a central value.* —**Brooks and Dick**
5. *The term dispersion is used to indicate the facts that within a given group, the items differ from one another in size or in other words, there is a lack of uniformity in their sizes.* —**W.I. King**

It is clear from these definitions that dispersion (also called scatter, spread or variation) measures the extent to which the individual items vary from a central value.

9.3 MEASURES OF DISPERSION

The degree to which numerical data tend to spread about an average value is called the **variation** or **dispersion** of the data. There are various measures of dispersion, the most important being the *range*, the *quartile deviation*, the *mean deviation* and the *standard deviation*. The measures of dispersion can be classified as follows:

(*i*) *Absolute measures of dispersion.*

(*ii*) *Relative measures of dispersion.*

Absolute Measures of Dispersion. The measures of dispersion which are expressed in the same statistical unit in which the original data are given are termed as **absolute measures of dispersion.** They have a serious drawback because the dispersion

obtained from these measures cannot be used to compare the variability of two distributions expressed in different units.

Relative Measures of Dispersion. These measures are pure numbers independent of the units of measurement and can be used to compare the variability of two distributions expressed in different units.

9.4 PROPERTIES OF A GOOD MEASURE OF DISPERSION

A good measure of dispersion should possess the following properties:

1. *It should be simple to understand.*
2. *It should be easy to calculate.*
3. *It should be rigidly defined.*
4. *It should be based on all the observations.*
5. *It should be amenable to further mathematical treatment.*
6. *It should have sampling stability.*
7. *It should not be unduly affected by extreme observations.*

9.5 RANGE

Definition. The **range** of a set of data is defined as the difference between the largest and the smallest value in the set. Symbolically,

$$\text{Range} = L - S$$

where L = largest value and S = smallest value.

In case of a grouped frequency distribution, range is defined as the difference between the upper limit of the highest class and the lower limit of the smallest class.

Coefficient of Range. The range is an absolute measure of dispersion and is expressed in the unit of measurement of values of a distribution. Hence it cannot be used to compare two distributions expressed in different units. To overcome this difficulty we need a relative measure which is independent of the units of measurement. This relative measure, called the **coefficient of range,** is defined as follows:

$$\text{Coefficient of range} = \frac{\text{Range}}{\text{Sum of the largest and the lowest values}} = \frac{L-S}{L+S}.$$

Example 1. Find the range and the coefficient of range for the following observations:

65, 70, 82, 59, 81, 76, 57, 60, 55 and 50

Solution. $\text{Range} = L - S$

Here, $L = 82$ and $S = 50$

$\therefore$ $\text{Range} = 82 - 50 = 32$

$$\text{Coefficient of range} = \frac{L-S}{L+S} = \frac{82-50}{82+50} = \frac{32}{132} = 0.24.$$

Example 2. Calculate range and coefficient of range from the following data:

Marks	:	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90
No. of Students	:	2	6	12	18	25	20	10	7

Solution. Range = $L - S = 90 - 10 = 80$

and Coefficient of range = $\frac{L-S}{L+S} = \frac{90-10}{90+10} = \frac{80}{100} = 0.8.$

Note. It should be noted that in the calculation of 'Range' only the values of the variable are taken into account and the frequencies are completely ignored.

9.6 MERITS AND DEMERITS OF RANGE

Merits. The range possesses the following merits:

1. *It is simple to understand and easy to calculate.*
2. *It requires minimum time to calculate the value of range.*

Demerits. The range has the following drawbacks:

1. *It is not based on all the observations.*
2. *Range is a poor measure of variation. It considers only the extreme values and tells us nothing about the distribution of numbers in between. Consider, for example, the following two sets of data, both with a range of 12:*

A	:	8	9	10	11	13	14	15	17	20
B	:	8	12	12	12	13	13	13	14	20

In set A the mean and median are both 13, but the numbers vary over the entire interval from 8 to 20. In set B the mean and median are also 13, but most of the values are closer to the center of the data.

3. *It is very much affected by fluctuations of sampling. Its value varies widely from sample to sample.*
4. *It cannot be calculated for grouped frequency distribution with open-end classes.*
5. *It is not suitable for further mathematical treatment.*

Uses of Range. Although the range is a poor measure of variation, it does have some useful applications in the following areas:

1. ***Industry.*** It is used in industry where the range for measurements on items coming off an assembly line might be specified in advance. As long as all measurements fall within the specified range, the process is said to be in control.
2. ***Weather Forecasts.*** It is used by the meteorological department for weather forecasts. The range in determining the difference between the minimum temperature and the maximum temperature is of great importance to the general public because they come to know the limits within which the temperature is likely to vary on a particular day.
3. ***Stock Exchange.*** It is useful in studying the fluctuations in the share prices.
4. ***Day-to-day Living.*** The range is a most widely used measure of dispersion in

our day-today life. For example, questions such as "What are the monthly wages of workers in the factory?" "What is the daily sales in a departmental store?" "How much do you travel in a month by your own car" are all answered in the form of range.

9.7 QUARTILE DEVIATION OR SEMI INTER-QUARTILE RANGE

Definition. **Inter-quartile range** is an absolute measure of dispersion defined by the formula:

$$\text{Inter-quartile range} = Q_3 - Q_1$$

where Q_1 and Q_3 are the first (or lower) and the third (or upper) quartiles respectively. As inter-quartile range is based only on Q_1 and Q_3, it measures the length of the interval that contains 50% of the data.

Definition. **Quartile deviation,** also called **semi inter-quartile range,** is an absolute measure of dispersion defined by the formula:

$$\text{Quartile Deviation } (Q.D.) = \frac{Q_3 - Q_1}{2}$$

Quartile deviation gives the average amount by which the two quartiles differ from the median. In a symmetrical distribution, the two quartiles Q_1 and Q_3 are equidistant from the median, *i.e.*, median $- Q_1 = Q_3 -$ median and as such median $\pm$ *Q.D.* covers exactly 50% of the data.

Quartile deviation, being an absolute measure of dispersion, cannot be used to compare two distributions expressed in different units. For comparative study of variability of two distributions, a relative measure known as *coefficient of quartile deviation* is defined.

Definition. The ***coefficient of quartile deviation*** is a relative measure of dispersion defined by

$$\text{Coefficient of Quartile Deviation} = \frac{\frac{Q_3 - Q_1}{2}}{\frac{Q_3 + Q_1}{2}} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Coefficient of quartile deviation is a pure number independent of the units of measurement and can be used to compare two distributions expressed in different units.

Example 3. Calculate quartile deviation from the following distribution :

X	:	5 – 7	8 – 10	11 – 13	14 – 16	17 – 19
Frequency	:	14	24	38	20	4

[Delhi Univ. B.Com. (P) 2001]

Solution. **Calculations For Quartiles**

Class Boundary	Frequency (f)	Less than c.f.
4.5 – 7.5	14	14
7.5 – 10.5	24	38
10.5 – 13.5	38	76
13.5 – 16.5	20	96
16.5 – 19.5	4	100
	$N = \sum f = 100$	

Computation of Q_1 : We have $\frac{N}{4} = \frac{100}{4} = 25$; and the c.f. just greater than or equal to 25 is 38. The class corresponding to this c.f. is 7.5 – 10.5. Thus Q_1– class is 7.5 – 10.5.

$$\therefore \quad Q_1 = l + \frac{\frac{N}{4} - C}{f} \times h = 7.5 + \frac{25 - 14}{24} \times 3 = 7.5 + 1.375 = 8.875$$

Computation of Q_3 : $\frac{3N}{4} = 75$; and the c.f. just greater than or equal to 75 is 76. The class corresponding to this c.f. is 10.5 – 13.5. Thus Q_3 – class is 10.5 – 13.5.

$$\therefore \quad Q_3 = l + \frac{\frac{3N}{4} - C}{f} \times h = 10.5 + \frac{75 - 38}{38} \times 3 = 10.5 + 2.92 = 13.42$$

Computation of Quartile Deviation:

$$Q.D. = \frac{Q_3 - Q_1}{2} = \frac{13.42 - 8.875}{2} = 2.27 \text{ (app.)}$$

Example 4. Find the value of third quartile if the values of first quartile and quartile deviation are 90 and 20 respectively. **[*Delhi Univ. B.Com. (P) 2013*]**

Solution. Quartile deviation is given by

$$Q.D. = \frac{Q_3 - Q_1}{2}$$

Substituting $Q_1 = 90$ and $Q.D. = 20$ in the above formula, we obtain $20 = \frac{Q_3 - 90}{2}$

$$\Rightarrow \quad Q_3 - 90 = 20 \times 2 = 40 \quad \Rightarrow \quad Q_3 = 90 + 40 = 130.$$

Example 5. If the first quartile is 48 and quartile deviation is 6, find the median (assuming the distribution to be symmetrical). **[*Delhi Univ. B.Com. (P) 1996*]**

Solution. Quartile deviation is given by

$$Q.D. = \frac{Q_3 - Q_1}{2}$$

Substituting $Q_1 = 48$ and $Q.D. = 6$ in the above formula, we obtain $6 = \frac{Q_3 - 48}{2}$

$$\Rightarrow \quad Q_3 - 48 = 12 \quad \Rightarrow \quad Q_3 = 60$$

For a symmetrical distribution, the first and the third quartiles are equidistant from the median. That is,

$$Q_3 - Md = Md - Q_1$$

or, $$2\,Md = Q_1 + Q_3 \Rightarrow Md = \frac{Q_1 + Q_3}{2}$$

$$\therefore \quad Md = \frac{48+60}{2} = \frac{108}{2} = 54.$$

Example 6. Find the inter-quartile range and the coefficient of quartile deviation from the following data:

Marks less than	:	10	20	30	40	50	60	70	80
No. of Students	:	4	16	40	76	96	112	120	125

Solution. We shall first convert the given cumulative frequency distribution into an ordinary frequency distribution.

Marks	*f*	*c.f.*
0 – 10	4	4
10 – 20	12	16
20 – 30	24	40
30 – 40	36	76
40 – 50	20	96
50 – 60	16	112
60 – 70	8	120
70 – 80	5	125
	$N = 125$	

Computation of Q_1 : $\frac{N}{4} = \frac{125}{4} = 31.25$; the c.f. just greater than or equal to 31.25 is 40. Therefore Q_1 lies in the class 20 – 30 and is given by

$$Q_1 = l + \frac{\frac{N}{4} - C}{f} \times h = 20 + \frac{31.25 - 16}{24} \times 10 = 26.35.$$

Computation of Q_3 : $\frac{3N}{4} = 93.75$; the *c.f.* just greater than or equal to 93.75 is 96. Therefore Q_3 lies in the class 40 – 50 and is given by

$$Q_3 = l + \frac{\frac{3N}{4} - C}{f} \times h = 40 + \frac{93.75 - 76}{20} \times 10 = 48.88$$

$\therefore$ Inter-quartile range $= Q_3 - Q_1 = 48.88 - 26.35 = 22.53$

and Coefficient of *Q.D.* $= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{48.88 - 26.35}{48.88 + 26.35} = \frac{22.53}{75.23} = 0.30$ (app.)

9.8 MERITS AND DEMERITS OF QUARTILE DEVIATION

Merits. 1. *It is simple to understand and easy to calculate.*

2. *It is superior to range as a measure of dispersion.*
3. *It can be computed from the distribution having open-end classes. In fact, it is the only measure of dispersion which can be obtained while dealing with a distribution having open-end classes.*
4. *Quartile deviation is not affected at all by extreme observations as it ignores 25% of the data from the beginning of the distribution and another 25% of the data from the top end.*
5. *Quartile deviation is useful specially when it is desired to study variability in the central half part of the data.*

Demerits. 1. *Quartile deviation is not based on all the observations. In fact, it ignores 25% of the data at the lower end and 25% of the data at the upper end. Hence it cannot be considered as a good measure of dispersion.*

2. *Quartile deviation is not suitable for further mathematical treatment.*
3. *It is affected considerably by sampling fluctuations.*

EXERCISE 9.1

Theory Questions

1. (*a*) What is meant by dispersion? **[*Delhi Univ. B.Com. (P) 1995, 2004*]**
   (*b*) List the various methods of measuring dispersion. **[*Delhi Univ. B.Com. (P) 1993, 2005, 2006*]**
2. What do you mean by relative measure of dispersion? **[*Delhi Univ. B.Com. (P) 2003*]**
3. Distinguish clearly between absolute and relative measures of dispersion. **[*Delhi Univ. B.Com. (P) 2008, 2012*]**
4. Explain briefly (*i*) Range, and (*ii*) Semi-interquartile range. **[*Delhi Univ. B.Com. (P) 1980*]**
5. What is coefficient of quartile deviation? State the formula. **[*Delhi Univ. B.Com. (P) 1994*]**

Practical Questions

1. Find the range of the daily wages of 10 persons given below:
   ₹240, ₹180, ₹ 250, ₹160, ₹200, ₹280, ₹220, ₹170, ₹210, ₹270
2. The following are the prices of shares of a company from Monday to Saturday:

Days :	*Monday*	*Tuesday*	*Wednesday*	*Thursday*	*Friday*	*Saturday*
Price (in ₹) :	55	54	52	53	56	58

   Calculate the range and the coefficient of range.
3. Find the range and the coefficient of range from the following data:

Diameter (mm) :	130	135	140	145	143	148	149	150
No. of Screws :	3	4	6	6	3	5	2	1

4. Calculate the coefficient of range from the following data:

Marks :	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
No. of Students :	16	18	20	17	9

5. Find the range and the coefficient of range of the weights of 100 male students of *ABC* University:

Weight (in kg)	:	60 – 62	63 – 65	66 – 68	69 – 71	72 – 74
No. of Students	:	5	18	42	27	8

6. If the first quartile is 142 and the semi-interquartile range is 18, find the median, (assuming the distribution to be symmetrical) **[*Delhi Univ. B.Com. (P) 1997*]**

7. Calculate the quartile deviation and its coefficient from the following data:

Month	:	1	2	3	4	5	6	7	8	9	10	11	12
Monthly Earnings (in ₹)	:	780	800	800	820	820	840	840	860	860	880	880	900

8. Calculate the quartile deviation and its coefficient from the following data :

Weight (in lbs)	:	120	122	124	126	130	140	150	160
No. of Students	:	1	3	5	7	10	3	1	1

9. Find (*i*) Inter-quartile range, (*ii*) Semi-interquartile range, (*iii*) Coefficient of quartile deviation from the following frequency distribution:

Marks	:	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90
No. of Students	:	60	45	120	25	90	80	120	60

10. Find the interquartile range and the coefficient of quartile deviation from the following data:

Size of item	:	16 – 20	21 – 25	26 – 30	31 – 35	36 – 40	41 – 45	46 – 50
Frequency	:	10	15	20	25	10	12	8

11. Find the value of third quartile if the values of first quartile and quartile deviation are 104 and 18 respectively. **[*Delhi Univ. B.Com. (P) 2002*]**

12. Find the value of third quartile if the values of first quartile and quartile deviation are 80 and 10 respectively. **[*Delhi Univ. B.Com. (P) 2010*]**

ANSWERS

1. ₹120 **2.** 6, 0.054 **3.** 20, 0.07 **4.** 0.714
5. 15, 0.11 **6.** 160 **7.** 35, 0.042 **8.** 3 lb, 0.0236
9. (*i*) 38.75; (*ii*) 19.375; (*iii*) 0.3647 **10.** 12.5, 0.2475 **11.** 140
12. 100

9.9 MEAN DEVIATION OR AVERAGE DEVIATION

As discussed earlier, the two measures of dispersion, *viz.*, range and quartile deviation are not based on all the observations. Moreover, they do not show any scatterness around an average and thus completely ignore the composition of the series. However, if we wish to measure variation in the sense of showing the scatterness around an average, we must include the deviations of each and every item from an average. *Mean deviation* or the *average deviation* helps us in achieving this goal. As the name suggests, this measure of dispersion is obtained on taking the average (arithmetic mean) of the deviations of the given values from a measure of central tendency. According to **Clark** and **Schkade:**

"*Average deviation is the average amount of scatter of the items in a distribution from either the mean or the median, ignoring the signs of the deviations. The average that is*

taken of the scatter is an arithmetic mean which accounts for the fact that this measure is often called the mean deviation".

Computation of Mean Deviation – Individual Observations

For a given set of n observations $X_1, X_2, \ldots, X_n$, the mean deviation *(M.D.)* about an average, say A, is given by

$$\text{Mean Deviation (about an average } A) = \frac{\sum |X - A|}{n} = \frac{\sum |D|}{n},$$

where $|D| = |X - A|$ (read as mod $(X - A)$ is the modulus value (or absolute value) of the deviation of X from A, ignoring $\pm$ signs.

Remark 1. Usually, we obtain mean deviation about any one of the three averages Mean (M), Median (Md) or Mode (Mo). But since mode is generally ill-defined, in practice, mean deviation is computed about mean or median. However, there is an advantage of computing mean deviation about median because *the sum of the deviations of items from median is least when signs are ignored.* Nevertheless, in practice, the mean is more frequently used in computing the average deviation and this is the reason why it is more commonly referred to as mean deviation.

Procedure for Computing the Mean Deviation

We now outline the procedure for computing the mean deviation:

Step 1. *Calculate the average A about which mean deviation is to be computed, by the methods discussed earlier.*

Step 2. *Find the deviation of each observation X from A and denote it by D. That is, find* $D = X - A$.

Step 3. *Find the absolute value of the deviation of each observation from A ignoring* $\pm$ *signs and denote it by* $|D|$.

Step 4. *Find the sum of all absolute deviations obtained in Step 3 to get* $\sum |D|$.

Step 5. *Divide the sum obtained in Step 4 by the number of observations to get the required mean deviation about the average A.*

Computation of Mean Deviation – Discrete Series

In case of discrete series where the variable X takes the values $X_1, X_2, \ldots, X_n$ with respective frequencies $f_1, f_2, \ldots, f_n$, the mean deviation about an average A is given by

$$\text{Mean Deviation (about an average } A) = \frac{\sum f|X - A|}{N} = \frac{\sum f|D|}{N},$$

where $N = \sum f$ is the total frequency and $D = X - A$

Procedure for Computing the Mean Deviation

Step 1. *Calculate the average A about which mean deviation is to be computed.*

Step 2. *Take the deviation of each observation from A and denote it by D. That is, find* $D = X - A$.

Step 3. *Find the absolute value of the deviation of each observation from A ignoring* $\pm$ *signs and denote it by* $|D|$.

Step 4. *Multiply each absolute deviation $|D|$ by the corresponding frequency f to get $f|D|$.*

Step 5. *Add all the products obtained in Step 4 to get $\sum f|D|$.*

Step 6. *Divide the sum obtained in Step 5 by N, the total frequency, to get the required mean deviation.*

Computation of Mean Deviation – Continuous Series

The computation of the mean deviation in the case of continuous series is exactly the same as discussed above for discrete series. The only difference is that here we have to obtain the class marks (or mid-values) of the various classes and take absolute deviations of these values from the average A. Thus, if $X_1, X_2, \ldots, X_n$ are the class marks (or mid-values) of a set of grouped data with corresponding class frequencies $f_1, f_2, \ldots, f_n$, then the mean deviation about an average A is given by

$$\text{Mean Deviation (about an average } A) = \frac{\sum f|X-A|}{N} = \frac{\sum f|D|}{N},$$

where $N = \sum f$ is the total frequency

Coefficient of Mean Deviation

The measures of mean deviation as defined above are absolute measures depending on the units of measurement. The relative measure corresponding to the mean deviation, called the **coefficient of mean deviation,** is given by

$$\text{Coefficient of } M.D. = \frac{\text{Mean Deviation}}{\text{Average about which it is calculated}}$$

$$\therefore \quad \text{Coefficient of } M.D. \text{ about mean} = \frac{M.D}{\text{Mean}}$$

$$\text{and} \quad \text{Coefficient of } M.D. \text{ about median} = \frac{M.D}{\text{Median}}$$

Coefficient of mean deviation is a pure number independent of the units of measurement and can be used to compare two distributions expressed in different units.

Example 7. Find the mean deviation about the median for the following data:

56 46 79 26 85 39 65 99 29 72

Find also the coefficient of mean deviation.

Solution. Arranging the data in ascending order of magnitude, we get

26 29 39 46 56 65 72 79 85 99

Here, $n = 10$ $\quad \therefore \quad \frac{n+1}{2} = \frac{10+1}{2} = 5.5$

$$\therefore \quad \text{Median} = \text{size of } \left(\frac{n+1}{2}\right)\text{th item} = \text{size of (5.5)th item}$$

$$= 5\text{th item} + 0.5\,(6\text{th item} - 5\text{th item})$$

$$= 56 + 0.5\,(65 - 56) = 56 + 4.5 = 60.5.$$

Calculation for Mean Deviation

X	*Deviation from Median* $D = X - Md$	*Absolute Deviation from Median* $\|D\|$
26	–34.5	34.5
29	–31.5	31.5
39	–21.5	21.5
46	–14.5	14.5
56	–4.5	4.5
65	4.5	4.5
72	11.5	11.5
79	18.5	18.5
85	24.5	24.5
99	38.5	38.5
$n = 10$		$\sum \|D\| = 204$

$$\therefore \quad M.D. \text{ (about median)} = \frac{\sum |D|}{n} = \frac{204}{10} = 20.4.$$

$$\text{Coefficient of } M.D. = \frac{\text{Mean Deviation}}{\text{Median}} = \frac{20.4}{60.5} = 0.34.$$

Example 8. Calculate mean deviation about the mean for the following data:

X :	10	11	12	13	14	Total
f :	3	12	18	12	3	48

Solution. **Calculations for Mean Deviation**

X	f	fX	$D = X - \bar{X}$	$\|D\|$	$f\|D\|$
10	3	30	– 2	2	6
11	12	132	– 1	1	12
12	18	216	0	0	0
13	12	156	1	1	12
14	3	42	2	2	6
	$N = \sum f = 48$	$\sum fX = 576$			$\sum f\|D\| = 36$

$$\text{Mean} : \bar{X} = \frac{\sum fX}{\sum f} = \frac{576}{48} = 12$$

$$\text{Mean Deviation about Mean} = \frac{\sum f\,|D|}{N} = \frac{36}{48} = 0.75.$$

Example 9. Calculate the mean deviation from the median for the following data:

Marks :	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90
No. of Students :	2	6	12	18	25	20	10	7

Also calculate the coefficient of mean deviation from median.

[Delhi Univ. B.Com. (P) 1982]

Solution. **Calculation for Mean Deviation**

Marks	Mid-value X	No. of Students f	Less than cf	D	$\lvert D\rvert$	$f\lvert D\rvert$
10 – 20	15	2	2	–39.8	39.8	79.6
20 – 30	25	6	8	–29.8	29.8	178.8
30 – 40	35	12	20	–19.8	19.8	237.6
40 – 50	45	18	38	–9.8	9.8	176.4
50 – 60	55	25	63	0.2	0.2	5.0
60 – 70	65	20	83	10.2	10.2	204.0
70 – 80	75	10	93	20.2	20.2	202.0
80 – 90	85	7	100	30.2	30.2	211.4
		$N = \sum f = 100$				$\sum f\lvert D\rvert = 1294.8$

Computation of Median : We have $\frac{N}{2} = 50$. The c.f. just greater than or equal to 50 is 63 and the corresponding class interval is 50-60. Thus the median class is 50 - 60.

$$\therefore \text{Median } (Md) = l + \frac{\frac{N}{2} - C}{f} \times h = 50 + \frac{50 - 38}{25} \times 10 = 50 + \frac{120}{25} = 50 + 4.8 = 54.8.$$

Computation of Mean Deviation :

$$M.D. = \frac{\sum f\,|D|}{N} = \frac{1294.8}{100} = 12.948.$$

$\therefore$ Coefficient of $M.D.$ from median

$$= \frac{M.D. \text{ (about median)}}{\text{Median}} = \frac{12.948}{54.8} = 0.24 \text{ (app.)}.$$

Example 10. Calculate the mean deviation from the mean for the following data:

Marks	:	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
No. of Students	:	6	5	8	15	7	6	3

Solution. **Calculation of Mean Deviation From Mean**

Marks	Mid-value X	No. of Students f	fX	$D = X - \bar{X}$	$\lvert D\rvert$	$f\lvert D\rvert$
0 – 10	5	6	30	–28.4	28.4	170.4
10 – 20	15	5	75	–18.4	18.4	92.0
20 – 30	25	8	200	–8.4	8.4	67.2
30 – 40	35	15	525	1.6	1.6	24.0
40 – 50	45	7	315	11.6	11.6	81.2
50 – 60	55	6	330	21.6	21.6	129.6
60 – 70	65	3	195	31.6	31.6	94.8
		$N = 50$	$\sum fX = 1670$			$\sum f\lvert D\rvert = 659.2$

Computation of Mean: $\bar{X} = \frac{\sum fX}{N} = \frac{1670}{50} = 33.4$

Computation of Mean Deviation about Mean:

$$M.D. = \frac{\sum f|D|}{N} = \frac{659.2}{50} = 13.18 \text{ (app.)}.$$

Example 11. Calculate the mean deviation and its coefficient:

Marks	: 21 – 25	26 – 30	31 – 35	36 – 40	41 – 45	46 – 50	51 – 55
No. of Students	: 5	15	28	42	15	12	3

[*Delhi Univ. B.Com. (P) 2004*]

Solution. The mean deviation gives the best results when deviations are taken from median. Since nothing is specified in the question, we shall take deviations from median.

Calculation of Mean Deviation

Marks	*Mid-Value X*	*No. of Students f*	*c.f.*	$D = X - Md$	$\|D\|$	$f\|D\|$
20.5 – 25.5	23	5	5	–13.93	13.93	69.65
25.5 – 30.5	28	15	20	– 8.93	8.93	133.95
30.5 – 35.5	33	28	48	–3.93	3.93	110.04
35.5 – 40.5	38	42	90	1.07	1.07	44.94
40.5 – 45.5	43	15	105	6.07	6.07	91.05
45.5 – 50.5	48	12	117	11.07	11.07	132.84
50.5 – 55.5	53	3	120	16.07	16.07	48.21
		$N = \sum f = 120$				$\sum f\|D\| = 630.08$

Computation of Median : $\frac{N}{2} = \frac{120}{2} = 60$; the *c.f.* just greater than or equal to 60 is 90.

Thus median lies in the class 35.5 - 40.5 and is given by

$$\text{Median } (Md) = l + \frac{\frac{N}{2} - C}{f} \times h = 35.5 + \frac{60 - 48}{42} \times 5 = 35.5 + 1.43 = 36.93$$

$$\therefore \quad M.D. \text{ (about Median)} = \frac{\sum f|D|}{N} = \frac{630.08}{120} = 5.26$$

$$\text{Coefficient of Mean Deviation (about median)} = \frac{M.D.}{\text{Median}} = 0.142.$$

9.10 MERITS AND DEMERITS OF MEAN DEVIATION

Merits. **1.** *It is easy to understand and simple to calculate.*

2. *It is based on each and every item of the data.*

3. *It is rigidly defined.*

4. As compared to standard deviation, it is less affected by extreme observations.
5. Since deviations are taken from a central value, comparison about formation of different distributions can easily be made.

Demerits. 1. The major drawback of mean deviation is that algebraic signs are ignored while taking the deviations of the items.

2. It is not suitable for further mathematical treatment.
3. It cannot be computed for distribution with open-end classes.
4. It is rarely used in sociological studies.

EXERCISE 9.2

Theory Questions

1. What do you understand by mean deviation?
2. What is coefficient of mean deviation? State the formula.

Practical Questions

1. Calculate the mean deviation about the median in respect of the following data:

   8 15 53 49 19 62 7 15 95 77

   [Calcutta Univ. B.Com. 1973]

2. Find the mean deviation about the mean in respect of the following data:

   31 35 29 63 55 72 37

3. Calculate the mean deviation about the mean for the following data:

X :	5	15	25	35	45	55	65
f :	8	12	10	8	3	2	7

4. Calculate the mean deviation about the median for the following data:

X :	2.5	3.5	4.5	5.5	6.5	7.5	8.5	9.5	10.5
f :	2	3	5	6	6	4	6	4	14

[Delhi Univ. B.Com. (P) 1981]

5. Calculate the mean deviation about the median for the following data:

Class Interval :	2 – 4	4 – 6	6 – 8	8 – 10
Frequency :	3	4	2	1

6. Find the mean deviation from the mean for the following data:

Class Interval :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
frequency :	8	12	10	8	3	2	7

7. Calculate the mean deviation from the mean for the following data and find its coefficient:

Marks :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of Students :	5	8	15	16	6

ANSWERS

1. 27.2 **2.** 14.86 **3.** 16 **4.** 2.22
5. 1.4 **6.** 16 **7.** *M.D.* = 9.44, Coefficient of *M.D.* = 0.35

9.11 STANDARD DEVIATION

The concept of standard deviation was first introduced by Karl Pearson in 1893. It is by far the most important and widely used measure of studying dispersion.

Definition. The **standard deviation,** abbreviated as *S.D.*, of a given set of observations is defined as the positive square root of the arithmetic mean of the squares of deviations of the observations from their arithmetic mean. It is denoted by the Greek small letter σ (read as sigma).

Thus standard deviation of a set of n observations $X_1, X_2, \ldots, X_n$ is given by

$$\sigma = \sqrt{\frac{(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \ldots + (X_n - \bar{X})^2}{n}} = \sqrt{\frac{\sum (X - \bar{X})^2}{n}},$$

where $\bar{X} = \frac{\sum X}{n}$ is the arithmetic mean of the given observations.

If $X_1, X_2, \ldots, X_n$ are the class marks of a set of grouped data with class frequencies $f_1, f_2, \ldots, f_n$, then the standard deviation is given by

$$\sigma = \sqrt{\frac{f_1(X_1 - \bar{X})^2 + f_2(X_2 - \bar{X})^2 + \ldots f_n(X_n - \bar{X})^2}{f_1 + f_2 + \ldots + f_n}} = \sqrt{\frac{\sum f(X - \bar{X})^2}{N}},$$

where $N = \sum f$ is the total frequency and $\bar{X} = \frac{\sum fX}{N}$ is the arithmetic mean.

Definition. The **variance** of a given set of observations is defined as the square of its standard deviation and is denoted by σ^2. Thus

$$\sigma^2 = \frac{\sum (X - \bar{X})^2}{n} \quad \text{or} \quad \sigma^2 = \frac{\sum f(X - \bar{X})^2}{N}$$

Example 12. Find variance if $\sum d^2 = 150$ and $N = 6$. Deviations are taken from actual mean. **[*Delhi Univ. B.Com. (P) 2010, 2011 (SOL)*]**

Solution. Variance $= \frac{\sum (X - \bar{X})^2}{N} = \frac{\sum d^2}{N} = \frac{150}{6} = 25.$

Example 13. Calculate standard deviation of the following marks obtained by 5 students in a tutorial group:

Marks obtained : 8 12 13 15 22 **[*Delhi Univ. B.Com. (P) 1997*]**

Solution. **Calculation of Standard Deviation**

Marks obtained (X)	$X - \bar{X}$	$(X - \bar{X})^2$
8	− 6	36
12	− 2	4
13	− 1	1
15	1	1
22	8	64
$\sum X = 70$		$\sum (X - \bar{X})^2 = 106$

$$\bar{X} = \frac{\sum X}{n} = \frac{70}{5} = 14$$

$\therefore$

$$\sigma = \sqrt{\frac{\sum (X - \bar{X})^2}{n}} = \sqrt{\frac{106}{5}} = \sqrt{21.2} = 4.6 \text{ (app.)}.$$

and

Example 14. Calculate the standard deviation for the following data:

X :	20	30	40	50	60	70
f :	8	12	20	10	6	4

Solution. **Calculation of Standard Deviation**

X	f	fX	$X - \bar{X}$	$(X - \bar{X})^2$	$f(X - \bar{X})^2$
20	8	160	–21	441	3528
30	12	360	–11	121	1452
40	20	800	–1	1	20
50	10	500	9	81	810
60	6	360	19	361	2166
70	4	280	29	841	3364
	$N = 60$	$\sum fX = 2460$			$\sum f(X - \bar{X})^2 = 11{,}340$

$\therefore$

$$\bar{X} = \frac{\sum fX}{N} = \frac{2460}{60} = 41$$

and

$$\sigma = \sqrt{\frac{\sum f(X - \bar{X})^2}{N}} = \sqrt{\frac{11340}{60}} = \sqrt{189} = 13.75.$$

Example 15. Calculate the mean and standard deviation from the following data:

Marks :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
No. of Students :	10	15	25	25	10	10	5

Solution. **Calculation of Standard Deviation**

Marks	*Mid value* X	f	*No. of Students* fX	$X - \bar{X}$	$(X - \bar{X})^2$	$f(X - \bar{X})^2$
0 – 10	5	10	50	–26	676	6760
10 – 20	15	15	225	–16	256	3840
20 – 30	25	25	625	–6	36	900
30 – 40	35	25	875	4	16	400
40 – 50	45	10	450	14	196	1960
50 – 60	55	10	550	24	576	5760
60 – 70	65	5	325	34	1156	5780
		$N = 100$	$\sum fX = 3100$			$\sum f(X - \bar{X})^2 = 25{,}400$

$$\therefore \quad \bar{X} = \frac{\sum fX}{N} = \frac{3100}{100} = 31$$

and $$\sigma = \sqrt{\frac{\sum f(X-\bar{X})^2}{N}} = \sqrt{\frac{25400}{100}} = \sqrt{254} = 15.94.$$

Different Methods of Calculating Standard Deviation – Ungrouped Data

By definition, the standard deviation of an ungrouped data consisting of n observations $X_1, X_2, \ldots, X_n$ is given by

$$\sigma = \sqrt{\frac{\sum(X-\bar{X})^2}{N}} \quad \text{where} \quad \bar{X} = \frac{\sum X}{n}$$

The computation of standard deviation by definition is very effective if $\bar{X}$ is an integer. However, if $\bar{X}$ comes out to be in fraction, its computation becomes very cumbersome and time-consuming. In that case we apply the following short-cut method which is very effective and reduces the numerical calculations to a great extent.

Formula 1 (Short-cut Method). According to this method, standard deviation is given by

$$\sigma = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2} = \sqrt{\frac{\sum X^2}{n} - \bar{X}^2}$$

Again if the values of X are large, Formula 1 can further be simplified by using what is known as an *assumed mean method.* When this method is used we take deviations of each item from an assumed mean, say A.

Formula 2 (Assumed Mean Method). According to this method, standard deviation is given by

$$\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}, \quad \text{where } d = X - A$$

Finally, if c is the common factor of the given data, then we may apply the following method, called *step-deviation method.*

Formula 3 (Step-deviation Method). According to this method, standard deviation is given by

$$\sigma = \sqrt{\frac{\sum u^2}{n} - \left(\frac{\sum u}{n}\right)^2} \times c, \quad \text{where } u = \frac{X-A}{c} \text{ and } c = \text{common factor.}$$

Remark. If we compare Formula 2 with Formula 1, we immediately conclude that standard deviation is independent of change of origin. However, if we compare Formula 3 with Formula 2, we find that standard deviation is not independent of change of scale. Thus the above two observations lead to the following important conclusion:

"Standard deviation of a distribution is independent of change of origin but not of scale"

Example 16. What will happen to standard deviation if all the items are increased by 2? Give reason. **[Delhi Univ. B.Com. (P) 2009]**

Solution. The standard deviation will not be affected if all the items are increased by 2, because it is independent of change of origin.

An Important Result: $\sum X^2 = n(\sigma^2 + \bar{X}^2)$

The above result follows immediately from Formula 1. In fact, applying Formula 1, we get

$$\sigma^2 = \frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2 = \frac{\sum X^2}{n} - \bar{X}^2$$

$\therefore$

$$\frac{1}{n}\sum X^2 = \sigma^2 + \bar{X}^2 \quad \text{or} \quad \sum X^2 = n(\sigma^2 + \bar{X}^2).$$

$\Rightarrow$

The above result is quite useful to find the correct value of standard deviation whenever one or more of the observations are wrongly copied down. For instance, *see Example 22.*

Different Methods of Calculating Standard Deviation – Grouped Data

If $X_1, X_2, \ldots, X_n$ are the class marks of a set of grouped data with corresponding class frequencies $f_1, f_2, \ldots, f_n$, then the standard deviation is given by

$$\sigma = \sqrt{\frac{\sum f(X - \bar{X})^2}{N}}$$

where $N = \sum f$ is the total frequency and $\bar{X} = \frac{\sum fX}{N}$ is the arithmetic mean of the distribution.

All the methods discussed earlier for calculating standard deviation in the case of ungrouped data can also be used in the case of grouped data. However, in practice it is the step deviation method that is mostly used.

Short-cut Method. According to this method, standard deviation is given by

$$\sigma = \sqrt{\frac{\sum fX^2}{N} - \left(\frac{\sum fX}{N}\right)^2}$$

Assumed Mean Method. According to this method, standard deviation is given by

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}, \quad \text{where } d = X - A$$

Step-deviation Method. According to this method, standard deviations is given by

$$\sigma = \sqrt{\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2} \times h, \quad \text{where } u = \frac{X - A}{h}$$

Example 17. From the following information, find standard deviations of X and Y variables:

$\sum X = 235, \quad \sum Y = 250, \quad \sum X^2 = 6750, \quad \sum Y^2 = 6840, \quad N = 10$

[Delhi Univ. B.Com. (H) 1997]

Solution. Standard Deviation of X variable.

$$\sigma_X = \sqrt{\frac{\sum X^2}{N} - \left(\frac{\sum X}{N}\right)^2} = \sqrt{\frac{6750}{10} - \left(\frac{235}{10}\right)^2} = \sqrt{675 - (23.5)^2} = \sqrt{675 - 552.25}$$

$$= \sqrt{122.75} = 11.08$$

Standard deviation of Y variable.

$$\sigma_Y = \sqrt{\frac{\sum Y^2}{N} - \left(\frac{\sum Y}{N}\right)^2} = \sqrt{\frac{6840}{10} - \left(\frac{250}{10}\right)^2} = \sqrt{684 - (25)^2} = \sqrt{684 - 625}$$

$$= \sqrt{59} = 7.68.$$

Example 18. From the following data, calculate the standard deviation :

X :	10	11	12	13	14	15	16	17	18
f :	2	7	10	12	15	11	10	6	3

[*Delhi Univ. B.Com. (P) 2005*]

Solution. **Calculation of Standard Deviation**

X	f	$d = X - 14$	fd	fd^2
10	2	−4	−8	32
11	7	−3	−21	63
12	10	−2	−20	40
13	12	−1	−12	12
14	15	0	0	0
15	11	1	11	11
16	10	2	20	40
17	6	3	18	54
18	3	4	12	48
	$N = 76$		$\sum fd = 0$	$\sum fd^2 = 300$

Standard deviation is given by

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} = \sqrt{\frac{300}{76}} = 1.986.$$

Example 19. Calculate the arithmetic mean and standard deviation from the following series:

Class Interval :	5 - 15	15 - 25	25 - 35	35 - 45	45 - 55
Frequency :	8	12	15	9	6

[*Delhi Univ. B.Com. (P) 1973*]

Solution. **Calculation of A.M. and S.D.**

Class Interval	Mid-value X	Frequency f	$u = \frac{X-A}{h}$ $(A = 30, h = 10)$	fu	fu^2
5 – 15	10	8	−2	−16	32
15 – 25	20	12	−1	−12	12
25 – 35	30	15	0	0	0
35 – 45	40	9	1	9	9
45 – 55	50	6	2	12	24
		$N = 50$		$\sum fu = -7$	$\sum fu^2 = 77$

Calculation of Arithmetic Mean: The *A.M.* is given by the formula

$$\bar{X} = A + \frac{\sum fu}{N} \times h = 30 + \frac{-7}{50} \times 10 = 30 - 1.4 = 28.6$$

Calculation of Standard Deviation: The S.D. is given by the formula

$$\sigma = \sqrt{\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2} \times h = \sqrt{\frac{77}{50} - \left(\frac{-7}{50}\right)^2} \times 10 = \sqrt{1.54 - 0.0196} \times 10$$

$$= 12.33.$$

Example 20. Find the mean and standard deviation of first n natural numbers.

[*Delhi Univ. B.Com. (P) 2014*]

Solution. The first n natural numbers are 1, 2, 3, ..., n. We know that

$$\sum X = 1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$$

and $$\sum X^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\therefore \quad \text{Mean} = \bar{X} = \frac{\sum X}{n} = \frac{1+2+3+\ldots+n}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2}$$

and $$S.D. = \sigma = \sqrt{\frac{\sum X^2}{n} - \bar{X}^2} = \sqrt{\frac{n(n+1)(2n+1)}{6n} - \left(\frac{n+1}{2}\right)^2}$$

$$= \sqrt{\frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}} = \sqrt{\frac{2(n+1)(2n+1) - 3(n+1)^2}{12}}$$

$$= \sqrt{\frac{(n+1)[4n+2-3n-3]}{12}} = \sqrt{\frac{(n+1)(n-1)}{12}} = \sqrt{\frac{n^2-1}{12}}$$

Thus, the mean of first n natural numbers is $\frac{n+1}{2}$ and their standard deviation is $\sqrt{\frac{n^2-1}{12}}$.

Example 21. The mean of 5 observations is 4.4 and the variance is 8.24. If three of the five observations are 1, 2 and 6, find the values of the other two.
[Delhi Univ. B.Com. (H) 1995, 2014 (SOL)]

Solution. We are given : $n = 5$, $\bar{X} = 4.4$ and $\sigma^2 = 8.24$

Now $\sum X = n\bar{X} = 5 \times 4.4 = 22$

and $\sum X^2 = n(\sigma^2 + \bar{X}^2) = 5\,(8.24 + 19.36) = 5 \times 27.60 = 138$

Let the two missing observations be X_1 and X_2. Then

$$\sum X = 22 \quad \Rightarrow \quad 1 + 2 + 6 + X_1 + X_2 = 22$$

$$\Rightarrow \quad X_1 + X_2 = 22 - 9 = 13 \qquad \ldots (1)$$

and $$\sum X^2 = 138 \quad \Rightarrow \quad 1^2 + 2^2 + 6^2 + X_1^2 + X_2^2 = 138$$

$$\Rightarrow \quad X_1^2 + X_2^2 = 138 - 41 = 97 \qquad \ldots (2)$$

Equation (1) gives $X_2 = 13 - X_1$. Substituting this value of X_2 in Equation (2), we obtain

$$X_1^2 + (13 - X_1)^2 = 97 \quad \Rightarrow \quad X_1^2 + 169 + X_1^2 - 26X_1 = 97$$

or, $$2X_1^2 - 26X_1 + 72 = 0 \quad \text{or} \quad X_1^2 - 13X_1 + 36 = 0$$

$$\Rightarrow \quad X_1 = \frac{13 \pm \sqrt{169 - 144}}{2} = \frac{13 \pm 5}{2} = 9 \quad \text{or} \quad 4$$

If $X_1 = 9$, then $X_2 = 4$, and if $X_1 = 4$, then $X_2 = 9$.
Thus the values of the other two observations are 9 and 4.

Correcting Incorrect values of Mean and Standard Deviation

Quite often while tabulating data some values are wrongly copied down. For example, it may happen that an observation, say 36, was wrongly copied down as 63 and we might have used this wrong value in the calculation of Mean and Standard Deviation. When such an error is found out there are two options – either to make all calculations a fresh, which is a very tedious task or to correct the values of mean and standard deviation by adjustment and readjustment of wrong and right figures. This second option is much easier and consumes much less time. To illustrate, let us consider the following example.

Example 22. The mean and variance of 100 items were worked out as 40 and 25 respectively by a student. By mistake an item 50 was wrongly taken as 5 in calculating the above. You are required to find correct mean and correct standard deviation.
[Delhi Univ. B.Com. (P) 2012]

Solution. In usual notations, we are given :

$$n = 100, \quad \bar{X} = 40 \quad \text{and} \quad \sigma^2 = 25$$

Now $\sum X = n\bar{X} = 100 \times 40 = 4000$

and $\sum X^2 = n(\sigma^2 + \bar{X}^2) = 100(25 + 1600) = 100(1625) = 1,62,500$

If the wrong observation 5 is replaced by the correct observation 50, then

$$\text{Corrected } \sum X = 4000 - 5 + 50 = 4045$$

$$\text{Corrected } \sum X^2 = 1,62,500 - 5^2 + (50)^2$$

$$= 1,62,500 - 25 + 2500 = 1,64,975$$

$$\therefore \quad \text{Corrected } \bar{X} = \frac{\text{Corrected } \sum X}{n} = \frac{4045}{100} = 40.45$$

$$\text{Corrected } \sigma = \sqrt{\frac{\text{Corrected } \sum X^2}{n} - \left(\text{Corrected } \bar{X}\right)^2}$$

$$= \sqrt{\frac{164975}{100} - (40.45)^2}$$

$$= \sqrt{1649.75 - 1636.20} = \sqrt{13.55} = 3.68$$

Example 23. The mean and standard deviation of 100 items are found to be 40 and 10. If at the time of calculations, two items are wrongly taken as 30 and 70 instead of 3 and 27, find the corrected mean and corrected standard deviation.

[*Delhi Univ. B.Com. (P) 2001, 2015*]

Solution. In usual notations, we are given : $n = 100$, $\bar{X} = 40$ and $\sigma = 10$

Now, $$\sum X = n\bar{X} = 100 \times 40 = 4000$$

and $$\sum X^2 = n\left(\sigma^2 + \bar{X}^2\right) = 100\,(100 + 1600) = 100\,(1700) = 1,70,000$$

If the wrong observations 30 and 70 are replaced by the correct observations 3 and 27, then

$$\text{Corrected } \sum X = 4000 - 30 - 70 + 3 \div 27 = 3930$$

$$\text{Corrected } \sum X^2 = 1,70,000 - (30)^2 - (70)^2 + (3)^2 + (27)^2$$

$$= 1,70,000 - 900 - 4900 + 9 + 729 = 1,64,938$$

$$\therefore \quad \text{Corrected } \bar{X} = \frac{\text{Corrected } \sum X}{n} = \frac{3930}{100} = 39.30$$

$$\text{Corrected } \sigma = \sqrt{\frac{\text{Corrected } \sum X^2}{n} - \left(\text{Corrected } \bar{X}\right)^2} = \sqrt{\frac{164938}{100} - (39.30)^2}$$

$$= \sqrt{1649.38 - 1544.49} = \sqrt{104.89} = 10.24.$$

9.12 COMBINED STANDARD DEVIATION

If two sets of data contain n_1 and n_2 observations having means $\bar{X}_1$ and $\bar{X}_2$, and standard deviations σ_1 and σ_2 respectively, then the standard deviation, σ, of the combined data with $n_1 + n_2$ observations is given by

$$\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}},$$

where $d_1 = \bar{X}_1 - \bar{X}$, $d_2 = \bar{X}_2 - \bar{X}$ and $\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$ is the combined mean.

The result can be generalised to more than two sets of data. For example, if $\bar{X}_1, \bar{X}_2, \ldots, \bar{X}_k$ be the means, $\sigma_1, \sigma_2, \ldots, \sigma_k$ be the standard deviations and $n_1, n_2, \ldots, n_k$ be the number of observations in each set, then the standard deviation of the combined data with $n_1 + n_2 + \ldots + n_k$ observations is given by

$$\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right) + n_k\left(\sigma_k^2 + d_k^2\right)}{n_1 + n_2 + \ldots + n_k}}.$$

where $d_1 = \bar{X}_1 - \bar{X}$, $d_2 = \bar{X}_2 - \bar{X}, \ldots,$ $d_k = \bar{X}_k - \bar{X}$

and $\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2 + \ldots + n_k\bar{X}_k}{n_1 + n_2 + \ldots + n_k}$ is the combined mean.

Example 24. From the following information, calculate the combined standard deviation:

$n_1 = 90$ $\bar{X}_1 = 20$ $\sigma_1 = 8$
$n_2 = 60$ $\bar{X}_2 = 15$ $\sigma_2 = 6$ **[*Delhi Univ. B.Com. (P) 2004*]**

Solution. Let $\bar{X}$ denote the combined mean and σ denote the combined standard deviation. Then

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2} = \frac{90 \times 20 + 60 \times 15}{90 + 60} = \frac{1800 + 900}{150} = \frac{2700}{150} = 18$$

and $$\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}},$$

where $d_1 = \bar{X}_1 - \bar{X} = 20 - 18 = 2 \Rightarrow d_1^2 = 4$

$d_2 = \bar{X}_2 - \bar{X} = 15 - 18 = -3 \Rightarrow d_2^2 = 9$

$$\therefore \quad \sigma = \sqrt{\frac{90(64 + 4) + 60(36 + 9)}{90 + 60}} = \sqrt{\frac{90 \times 68 + 60 \times 45}{150}}$$

$$= \sqrt{\frac{6120 + 2700}{150}} = \sqrt{\frac{8820}{150}} = \sqrt{58.8} = 7.67 \text{ (app.)}$$

Example 25. Find the missing information from the following :

	Group I	Group II	Group III	Combined
Number	50	?	90	200
Standard Deviation	6	7	?	7.746
Mean	113	?	115	116

Solution. Let n_1, n_2, n_3 denote the number of observations in Group *I*, Group *II*, Group *III* respectively. Then we are given **[Delhi Univ. B.Com. (H) 1992]**

$$n_1 + n_2 + n_3 = 200$$

But $n_1 = 50$ and $n_3 = 90$ $\therefore$ $50 + n_2 + 90 = 200 \Rightarrow n_2 = 60$

Let $\bar{X}_1, \bar{X}_2, \bar{X}_3$ denote the means of Group *I*, Group *II*, Group *III* respectively. Then the combined mean, $\bar{X}$, is given by

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2 + n_3\bar{X}_3}{n_1 + n_2 + n_3} \Rightarrow 116 = \frac{50 \times 113 + 60 \times \bar{X}_2 + 90 \times 115}{200}$$

$$\Rightarrow 23200 = 5650 + 60\,\bar{X}_2 + 10350 \Rightarrow 60\,\bar{X}_2 = 23200 - 5650 - 10350 = 7200$$

$$\Rightarrow \bar{X}_2 = \frac{7200}{60} = 120$$

Let $\sigma_1, \sigma_2, \sigma_3$ denote the standard deviations of Group *I*, Group *II*, Group *III* respectively. Then the combined standard deviation, σ, is given by

$$\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right) + n_3\left(\sigma_3^2 + d_3^2\right)}{n_1 + n_2 + n_3}}$$

$$\Rightarrow (n_1 + n_2 + n_3)\sigma^2 = n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2) + n_3(\sigma_3^2 + d_3^2)$$

where $d_1 = \bar{X}_1 - \bar{X} = 113 - 116 = -3$

$d_2 = \bar{X}_2 - \bar{X} = 120 - 116 = 4$

$d_3 = \bar{X}_3 - \bar{X} = 115 - 116 = -1$

$\therefore$ $200\,(7.746)^2 = 50\,(36 + 9) + 60\,(49 + 16) + 90\,(\sigma_3^2 + 1)$

$\Rightarrow$ $200\,(60.000516) = 2250 + 3900 + 90 + 90\,\sigma_3^2$

$\Rightarrow$ $12000 = 6240 + 90\,\sigma_3^2$

$\Rightarrow$ $90\,\sigma_3^2 = 12000 - 6240 = 5760$

$\Rightarrow$ $\sigma_3^2 = \frac{5760}{90} = 64 \Rightarrow \sigma_3 = 8$ (rejecting – ve value)

Thus the missing figures are : $n_2 = 60$, $\bar{X}_2 = 120$ and $\sigma_3 = 8$.

Example 26. The first of two sub-groups has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, find the mean and standard deviation of the second sub-group.

Solution. The given information can be put in the tabular form as follows :

	Sub group I	Sub group II	Combined
No. of items	$n_1 = 100$	$n_2 = 150$	$n_1 + n_2 = 250$
Mean	$\bar{X}_1 = 15$	$\bar{X}_2 = ?$	$\bar{X} = 15.6$
Standard deviation	$\sigma_1 = 3$	$\sigma_2 = ?$	$\sigma = \sqrt{13.44}$

Applying the formula for the combined mean :

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2} \quad \text{we get} \quad 15.6 = \frac{100 \times 15 + 150\bar{X}_2}{250}$$

$$\Rightarrow \quad 3900 = 1500 + 150\,\bar{X}_2 \quad \Rightarrow \quad 150\,\bar{X}_2 = 2400 \quad \Rightarrow \quad \bar{X}_2 = 16$$

Also, $$\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}},$$

where $d_1 = \bar{X}_1 - \bar{X} = 15 - 15.6 = -0.6$

$d_2 = \bar{X}_2 - \bar{X} = 16 - 15.6 = 0.4$

$$\therefore \quad 13.44 = \frac{100(9 + 0.36) + 150\left(\sigma_2^2 + 0.16\right)}{250}$$

$$\Rightarrow 13.44 \times 250 = 936 + 24 + 150\,\sigma_2^2$$

or, $3360 = 960 + 150\,\sigma_2^2 \Rightarrow 150\,\sigma_2^2 = 2400 \Rightarrow \sigma_2^2 = 16$ or $\sigma_2 = 4$

Thus the mean and standard deviation of the second sub-group are 16 and 4 respectively.

Example 27. For a group containing 100 items, the arithmetic mean and standard deviation are 8 and $\sqrt{10.5}$. For 50 observations selected from these 100 observations, the mean and standard deviation are 10 and 2 respectively. Find the mean and standard deviation of the remaining 50 observations. **[*Delhi Univ. B.Com (P) 2008 (SOL)*]**

Solution. The given information can be put in the tabular form as follows :

	Group I	Group II	Combined
Number	$n_1 = 50$	$n_2 = 50$	$n_1 + n_2 = 100$
Mean	$\bar{X}_1 = 10$	$\bar{X}_2 = ?$	$\bar{X} = 8$
Standard deviation	$\sigma_1 = 2$	$\sigma_2 = ?$	$\sigma = \sqrt{10.5}$

We have $$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2} \quad \Rightarrow \quad 8 = \frac{50 \times 10 + 50\bar{X}_2}{100}$$

$$\Rightarrow \quad 800 = 500 + 50\,\bar{X}_2 \quad \Rightarrow \quad 50\,\bar{X}_2 = 300 \quad \Rightarrow \quad \bar{X}_2 = 6$$

Also $$\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}}$$

$(n_1 + n_2)\,\sigma^2 = n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)$,

where $d_1 = \bar{X}_1 - \bar{X} = 10 - 8 = 2$ and $d_2 = \bar{X}_2 - \bar{X} = 6 - 8 = -2$

$\therefore \quad 100\,(10.5) = 50\,(4 + 4) + 50\,(\sigma_2^2 + 4) \Rightarrow 1050 = 400 + 200 + 50\sigma_2^2$

$\Rightarrow \quad 50\sigma_2^2 = 1050 - 600 = 450 \Rightarrow \sigma_2^2 = \frac{450}{50} = 9$ or $\sigma_2 = 3$

Thus the mean and standard deviation of the remaining 50 items are 6 and 3 respectively.

Example 28. For two groups of observations the following results were available:

Group I	*Group II*
$\sum(X-5) = 8$	$\sum(X-8) = -10$
$\sum(X-5)^2 = 40$	$\sum(X-8)^2 = 70$
$n_1 = 20$	$n_2 = 25$

Find the mean and standard deviation of both the groups taken together.

[Delhi Univ B.Com (P) 2011 (SOL)]

Solution. In terms of usual notations, we are given the following information :

	Group I	*Group II*
$\sum d$	8	− 10
$\sum d^2$	40	70
n	20	25
A	5	8

Computation of Mean and Standard Deviation

Group I	*Group II*
$\bar{X}_1 = A + \frac{\sum d}{n} = 5 + \frac{8}{20} = 5 + 0.4 = 5.4$	$\bar{X}_2 = A + \frac{\sum d}{n} = 8 + \frac{-10}{25} = 8 - 0.4 = 7.6$
$\sigma_1 = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2} = \sqrt{\frac{40}{20} - \left(\frac{8}{20}\right)^2}$	$\sigma_2 = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2} = \sqrt{\frac{70}{25} - \left(\frac{-10}{25}\right)^2}$
$= \sqrt{2 - 0.16} = \sqrt{1.84} = 1.36$	$= \sqrt{2.8 - 0.16} = \sqrt{2.64} = 1.62$

Computation of Combined Mean and Combined Standard Deviation

Combined Mean : $\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2} = \frac{20 \times 5.4 + 25 \times 7.6}{20 + 25} = \frac{108 + 190}{45} = \frac{298}{45} = 6.62.$

Combined S.D. : $\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}}$,

where $d_1 = \bar{X}_1 - \bar{X} = 5.4 - 6.62 = -1.22 \Rightarrow d_1^2 = 1.48$ (app.)

$$d_2 = \bar{X}_2 - \bar{X} = 7.6 - 6.62 = 0.98 \quad \Rightarrow \quad d_2^2 = 0.96 \text{ (app.)}$$

$$\therefore \quad \sigma = \sqrt{\frac{20(1.84+1.48)+25(2.62+0.96)}{20+25}} = \sqrt{\frac{20\times 3.32+25\times 3.58}{45}}$$

$$= \sqrt{\frac{66.4+89.5}{45}} = \sqrt{\frac{155.9}{45}} = \sqrt{3.46} = 1.86 \text{ (app.)}.$$

Example 29. The mean height of students in a class is 152 cm. The mean height of boys is 158 cms. with a standard deviation of 5 cm and the mean height of girls is 148 cm with a standard deviation of 4 cm. Find the percentage of boys and girls in the class and the standard deviation of all the students of the class taken together.

[*Delhi Univ. B.Com. (H) 2013*]

Solution. The given information can be put in the tabular form as follows :

	Boys	*Girls*	*Total*
Number	n_1	n_2	$n_1 + n_2$
Mean height	$\bar{X}_1 = 158$	$\bar{X}_2 = 148$	$\bar{X} = 152$
Standard deviation	$\sigma_1 = 5$	$\sigma_2 = 4$	$\sigma = ?$

Applying the formula $\bar{X} = \dfrac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$ for the combined mean, we get

$$152 = \frac{158n_1 + 148n_2}{n_1 + n_2} \quad \Rightarrow \quad 152n_1 + 152n_2 = 158n_1 + 148n_2$$

$$\Rightarrow \quad 4n_2 = 6n_1 \quad \Rightarrow \quad n_2 = 1.5\, n_1 \quad \text{or} \quad \frac{n_1}{n_2} = \frac{2}{3}$$

$\Rightarrow$ Number of boys and girls are in the ratio 2 : 3

$$\therefore \text{ \% of boys} = \frac{2}{2+3}\times 100 = 40\% \quad \text{and \% of girls} = \frac{3}{2+3}\times 100 = 60\%$$

The combined standard deviation of all the students is given by

$$\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}}$$

where, $d_1 = \bar{X}_1 - \bar{X} = 158 - 152 = 6 \quad \Rightarrow \quad d_1^2 = 36$

$$d_2 = \bar{X}_2 - \bar{X} = 148 - 152 = -4 \quad \Rightarrow \quad d_2^2 = 16$$

$$\therefore \quad \sigma = \sqrt{\frac{n_1(25+36)+n_2(16+16)}{n_1+n_2}} = \sqrt{\frac{61n_1+32n_2}{n_1+n_2}} = \sqrt{\frac{61n_1+32(1.5)n_1}{n_1+1.5n_1}}$$

$(\because n_2 = 1.5n_1)$

$$= \sqrt{\frac{61n_1+48n_1}{2.5n_1}} = \sqrt{\frac{109n_1}{2.5n_1}} = \sqrt{\frac{109}{2.5}} = \sqrt{43.6} = 6.60 \text{ cm.}$$

9.13 COEFFICIENT OF VARIATION

The standard deviation is an absolute measure of dispersion, depending upon the units of measurement. It does not tell us much about the variability of a single set of data. The *coefficient of standard deviation*, based on standard deviation, is a relative measure of dispersion and is given by

$$\text{Coefficient of Standard Deviation} = \frac{S.D.}{\text{Mean}} = \frac{\sigma}{\bar{X}}$$

This is a pure number independent of the units of measurement and hence can be used to compare the variability of two distributions expressed in different units. Perhaps a more appropriate measure is the *coefficient of variation* (*C.V.*), defined by

$$\text{Coefficient of Variation} = \frac{S.D.}{\text{Mean}} \times 100 = \frac{\sigma}{\bar{X}} \times 100$$

which expresses the standard deviation as a percentage of the mean. Since *C.V.* is a measure of *relative variation* expressed as a percent, the coefficient of variation can be used to compare the variability of two or more sets of data even when the observations are expressed in different units of measurement.

A distribution for which the coefficient of variation is smaller is said to be less variable or more consistent, more uniform, more stable or more homogeneous. On the other hand, the distribution for which the coefficient of variation is greater is said to be more variable or less consistent, less uniform, less stable or less homogeneous.

Example 30. The algebraic sum of 10 items about 8 is –5. Find its arithmetic mean. Also find coefficient of variation if standard deviation is 1.5.

[*Delhi Univ. B.Com. (P) 2014 (SOL)*]

Solution. In terms of usual notations, we are given :

$$n = 10, \quad \sum d = -5 \quad \text{and} \quad A = 8$$

$$\therefore \quad \bar{X} = A + \frac{\sum d}{n} = 8 + \frac{-5}{10} = 7.5$$

If $S.D. = 1.5$, the coefficient of variation is given by

$$C.V. = \frac{S.D.}{\bar{X}} \times 100 = \frac{1.5}{7.5} \times 100 = 20\%.$$

Example 31. The mean and standard deviation of two brands of light bulbs are given below :

	Brand I	*Brand II*
Mean	800 hours	770 hours
Standard deviation	100 hours	60 hours

Calculate a measure of relative dispersion for the two brands and interpret the result.

[*Delhi Univ B.Com. (H) 2000*]

Solution. Coefficient of variation for Brand $I = \frac{\sigma}{\bar{X}} \times 100 = \frac{100}{800} \times 100 = 12.5\%$

Coefficient of variation for Brand $II = \frac{\sigma}{\bar{X}} \times 100 = \frac{60}{770} \times 100 = 7.8\%$.

Since C.V. for Brand *II* is less than C.V. for Brand *I*, Brand *II* bulbs are considered to be more consistent in performance.

Example 32. The weekly wages of workers in a factory increases from ₹800 to ₹1200 and and the standard deviation increases from ₹100 to ₹500. Have the wages become less uniform now? **[*Delhi Univ. B.Com.(P) 2012*]**

Solution. CV (before increase) $= \frac{\sigma}{\bar{X}} \times 100 = \frac{100}{800} \times 100 = 12.5\%$

CV (after increase) $= \frac{\sigma}{\bar{X}} \times 100 = \frac{500}{1200} \times 100 = 41.67\%$

$\because$ CV (before increase) is less than CV (after increase), therefore, wages have become less uniform now.

Example 33. "After settlement the average weekly wage in a factory increased from ₹8000 to ₹12000 and standard deviation had increased from ₹100 to ₹150. After settlement the wage has become higher and more uniform." Do you agree? **[*Delhi Univ B.Com. (H) 2005*]**

Solution. Since the average weekly wage has increased from ₹8000 to ₹12000, therefore weekly wages have become higher.

$$C.V. \text{ (before settlement)} = \frac{100}{8000} \times 100 = 1.25\%$$

$$C.V. \text{ (after settlement)} = \frac{150}{12000} \times 100 = 1.25\%$$

Since there is no change in $C.V.$, there is no improvement in uniformity.

Example 34. Average runs scored by three batsman A, B and C are 50, 48 and 12 respectively. Their standard deviations of runs scored are 15, 12 and 2 respectively.

(*i*) Who is the highest run getter assuming each one of them have played equal number of innings?

(*ii*) If one of them is to be selected on the basis of consistency, who will be selected? **[*Delhi Univ. B.Com. (P) 2009*]**

Solution. Suppose each batsman has played n innings. Then

Total runs scored by batsman $A = 50n$ $\quad (\because \sum X = n\bar{X})$

Total runs scored by batsman $B = 48n$

Total runs scored by batsman $C = 12n$

Comparing the above scores, we conclude that batsman A is the highest run getter.

(*ii*) Now $C.V.$ (batsman A) $= \frac{15}{50} \times 100 = 30\% \quad \left(\because C.V = \frac{S.D.}{Mean} \times 100\right)$

$C.V.$ (batsman B) $= \frac{12}{48} \times 100 = 25\%$

$$C.V.\text{ (batsman C)} = \frac{2}{12} \times 100 = 16.67\%$$

Since *C.V.* (batsman C) is least, batsman C is considered to be most consistent and consequently he should be selected on the basis of consistency.

Example 35. The following is the record of goals scored by team '*A*' in a football season:

Number of goals scored	:	0	1	2	3	4
Number of matches played	:	1	9	7	5	3

For team '*B*', the average number of goals scored per match was 2.5 with a standard deviation of 1.25 goals. Find which team may be considered more consistent?

[*Delhi Univ. B.Com. (P) 2006*]

Solution. **Calculation of A.M. and S.D. of Team *A***

No. of goals (X)	*No. of matches (f)*	fX	fX^2
0	1	0	0
1	9	9	9
2	7	14	28
3	5	15	45
4	3	12	48
	$N = 25$	$\sum fX = 50$	$\sum fX^2 = 130$

$$\therefore \quad \bar{X} = \frac{\sum fX}{N} = \frac{50}{25} = 2$$

$$\text{and } S.D. = \sigma = \sqrt{\frac{\sum fX^2}{N} - \left(\frac{\sum fX}{N}\right)^2} = \sqrt{\frac{130}{25} - \left(\frac{50}{25}\right)^2} = \sqrt{5.2 - 4} = \sqrt{1.2} = 1.09$$

$$\text{Coefficient of variation for Team } A = \frac{S.D.}{\text{Mean}} \times 100 = \frac{1.09}{2} \times 100 = 54.5\,\%$$

$$\text{Coefficient of variation for Team } B = \frac{S.D.}{\text{Mean}} \times 100 = \frac{1.25}{2.5} \times 100 = 50\%$$

Since *C.V.* for Team *B* is less than *C.V.* for Team *A*, Team *B* is considered to be more consistent.

Example 36. Calculate standard deviation and coefficient of variation from the data given below:

Mid-point	:	5	15	25	35	45	55	65	75
Frequency	:	5	8	7	12	28	20	10	10

[*Delhi Univ. B.Com. (P) 1980*]

Solution. Calculation of Standard Deviation and Coefficient of Variation

X	$u = \frac{X - 45}{10}$ $(A = 45, h = 10)$	f	fu	fu^2
5	– 4	5	– 20	80
15	– 3	8	– 24	72
25	– 2	7	– 14	28
35	– 1	12	– 12	12
45	0	28	0	0
55	+ 1	20	+ 20	20
65	+ 2	10	+ 20	40
75	+ 3	10	+ 30	90
		$N = 100$	$\sum fu = 0$	$\sum fu^2 = 342$

Computation of arithmetic mean

$$\bar{X} = A + \frac{\sum fu}{N} \times h = 45 + \frac{0}{100} \times 10 = 45$$

Computation of standard deviation

$$\sigma = \sqrt{\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2} \times h = \sqrt{\frac{342}{100} - \left(\frac{0}{100}\right)^2} \times 10$$

$$= 1.849 \times 10 = 18.49$$

Computation of coefficient of variation

$$C.V. = \frac{\sigma}{\bar{X}} \times 100 = \frac{18.49}{45} \times 100 = 41.09\%.$$

Example 37. A charitable organisation decided to give old age pension to people over sixty years of age. The scales of pension were fixed as follows :

Age Group (years)	*Amount of Pension (₹)*
60 – 65	200 per month
65 – 70	250 per month
70 – 75	300 per month
75 – 80	350 per month
80 – 85	400 per month

The ages of 25 workers who secured the pension rights are given as follows :

74	62	84	72	61	83	72	81	64	71	63	61
60	67	74	64	79	75	76	69	68	78	66	67

Calculate the monthly average pension payable, its standard deviation and coefficient of variation.

[*Delhi Univ. B.Com. (P) 2015*]

Solution. Calculation of Average Monthly Pension and Standard Deviation

Age Group (years)	Tally Marks	Frequency (f)	Amt. of Pension (X)	$u = \frac{X-300}{50}$	fu	fu^2
60 – 65	|||| || 7	200	– 2	– 14	28
65 – 70	||||	5	250	– 1	– 5	5
70 – 75	||||	6	300	0	0	0
75 – 80	||||	4	350	1	4	4
80 – 85	|||	3	400	2	6	12
		$N = \sum f$ = 25			$\sum fu$ = – 9	$\sum fu^2$ = 49

Computation of Average Monthly Pension

The average monthly pension is given by

$$\bar{X} = A + \frac{\sum fu}{N} \times h = 300 + \frac{-9}{25} \times 50 = ₹282$$

Computation of Standard Deviation :

$$\sigma = \sqrt{\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2} \times h = \sqrt{\frac{49}{25} - \left(\frac{-9}{25}\right)^2} \times 50$$

$$= \sqrt{1.96 - 0.1296} \times 50 = \sqrt{1.8304} \times 50 = 1.3529 \times 50 = ₹67.64.$$

$$C.V. = \frac{\sigma}{\bar{X}} \times 100 = \frac{67.64}{282} \times 100 = 23.98\%.$$

Example 38. A purchasing agent obtained samples of lamps from two suppliers. He had the samples tested in his own laboratory for the lengths of life with the following results:

Length of life (in hours)	Samples from	
	Company A	Company B
700 – 900	10	3
900 – 1100	16	42
1100 – 1300	26	12
1300 – 1500	8	3

(i) Which company's lamps have greater average life?

(ii) Which company's lamps are more uniform? **[Delhi Univ. B.Com. (P) 2006]**

Solution. **Calculations for Mean and Standard Deviation**

Length of life (in hours)	Mid-value X	$u = \frac{X - 1000}{200}$ ($A = 1000$, $h = 200$)	Company A			Company B		
			f	fu	fu^2	f	fu	fu^2
700 – 900	800	– 1	10	– 10	10	3	–3	3
900 –1100	1000	0	16	0	0	42	0	0
1100 –1300	1200	1	26	26	26	12	12	12
1300 –1500	1400	2	8	16	32	3	6	12
			60	32	68	60	15	27

Company A :

$$\bar{X} = A + \frac{\sum fu}{N} \times h$$

$$= 1000 + \frac{32}{60} \times 200$$

$$= 1000 + 106.67 = 1106.67$$

$$\sigma = \sqrt{\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2} \times h$$

$$= \sqrt{\frac{68}{60} - \left(\frac{32}{60}\right)^2} \times 200$$

$$= \sqrt{1.133 - 0.284} \times 200$$

$$= \sqrt{0.849} \times 200 = 0.921 \times 200$$

$$= 184.2$$

$$C.V. = \frac{\sigma}{\bar{X}} \times 100 = \frac{184.2}{1106.67}$$

$$= 16.64\,\%$$

Company B :

$$\bar{X} = A + \frac{\sum fu}{N} \times h$$

$$= 1000 + \frac{15}{60} \times 200$$

$$= 1000 + 50 = 1050$$

$$\sigma = \sqrt{\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2} \times h$$

$$= \sqrt{\frac{27}{60} - \left(\frac{15}{60}\right)^2} \times 200$$

$$= \sqrt{0.45 - 0.0625} \times 200$$

$$= \sqrt{0.3875} \times 200 = 0.622 \times 200$$

$$= 124.4$$

$$C.V. = \frac{\sigma}{\bar{X}} \times 100 = \frac{124.4}{1050} \times 100$$

$$= 11.85\,\%$$

(*i*) The lamps of company *A* have greater average life.

(*ii*) The lamps of company *B* are more uniform, because *C.V.* of samples of bulbs from company *B* is less than that of company *A*.

Example 39. The following data gives the distribution of marks of 150 students of section *A* and 100 students of section *B*, the students of which section show the consistency in getting the marks: **[*Delhi Univ. B.Com. (P) 2012*]**

Marks	No. of Students (Sec A)	No. of Students (Sec B)
Less than 15	17	24
Less than 30	68	50
Less than 45	130	90
Less than 60	142	95
Less than 75	150	100

Solution. The data is first converted into an ordinary frequency distribution as shown below:

Calculation for Mean and Standard Deviation

Marks	Mid-value X	$u = \frac{X - 37.5}{15}$, $A = 37.5$, $h = 15$	Section A			Section B		
			f	fu	fu^2	f	fu	fu^2
0 - 15	7.5	−2	17	−34	68	24	− 48	96
15 - 30	22.5	−1	68 - 17 = 51	−51	51	50 − 24 = 26	− 26	26
30 - 45	37.5	0	130 - 68 = 62	0	0	90 − 50 = 40	0	0
45 - 60	52.5	1	142 - 130 = 12	12	12	95 − 90 = 5	5	5
60 - 75	67.5	2	150 − 142 = 8	16	32	100 − 95 = 5	10	20
			$N = \Sigma f = 150$	$\Sigma fu = -57$	$\Sigma fu^2 = 163$	$N = \Sigma f = 100$	$\Sigma fu = -59$	$\Sigma fu^2 = 147$

Section A

$$\bar{X} = A + \frac{\Sigma fu}{N} \times h = 37.5 + \frac{-57}{150} \times 15$$

$$= 37.5 - 5.7 = 31.8$$

$$\sigma = \sqrt{\frac{\Sigma fu^2}{N} - \left(\frac{\Sigma fu}{N}\right)^2} \times h$$

$$= \sqrt{\frac{163}{150} - \left(\frac{-57}{150}\right)^2} \times 15$$

$$= \sqrt{1.087 - 0.144} \times 15$$

$$= 0.971 \times 15 = 14.57$$

$$C.V. = \frac{\sigma}{\bar{X}} \times 100$$

$$= \frac{14.57}{31.8} \times 100 = 45.82\%$$

Section B

$$\bar{X} = A + \frac{\Sigma fu}{N} \times h = 37.5 + \frac{-59}{100} \times 15$$

$$= 37.5 - 8.85 = 28.65$$

$$\sigma = \sqrt{\frac{\Sigma fu^2}{N} - \left(\frac{\Sigma fu}{N}\right)^2} \times h$$

$$= \sqrt{\frac{147}{100} - \left(\frac{-59}{100}\right)^2} \times 15$$

$$= \sqrt{1.47 - 0.348} \times 15$$

$$= 1.059 \times 15 = 15.89$$

$$C.V. = \frac{\sigma}{\bar{X}} \times 100$$

$$= \frac{15.89}{28.65} \times 100 = 55.46\%$$

∵ C.V. (*Section A*) is less than C.V. (*Section B*), the students of Section A show more consistency in getting the marks.

Example 40. Share prices of two companies *A* Ltd., and *B* Ltd. were recorded as follows :

A Ltd. :	12	13	15	14	14	14	13	17
B Ltd. :	113	114	113	115	117	114	112	114

Which company's share prices are more variable ? **[*Delhi Univ. B.Com. (P) 2005*]**

Solution.

Share Price of A X	$A = 14$ $d_X = X - 14$	d_X^2	Share Price of B Y	$B = 114$ $d_Y = Y - 114$	d_Y^2
12	− 2	4	113	− 1	1
13	− 1	1	114	0	0
15	1	1	113	− 1	1
14	0	0	115	1	1
14	0	0	117	3	9
14	0	0	114	0	0
13	− 1	1	112	− 2	4
17	3	9	114	0	0
$n = 8$	$\sum d_X = 0$	$\sum d_X^2 = 16$	$n = 8$	$\sum d_Y = 0$	$\sum d_Y^2 = 16$

Company A Ltd. :

$$\bar{X} = A + \frac{\sum d_X}{n}$$

$$= 14 + \frac{0}{8} = 14$$

$$\sigma_X = \sqrt{\frac{\sum d_X^2}{n} - \left(\frac{\sum d_X}{n}\right)^2}$$

$$= \sqrt{\frac{16}{8}} = \sqrt{2} = 1.414$$

$$C.V.\,(X) = \frac{\sigma_X}{\bar{X}} \times 100$$

$$= \frac{1.414}{14} \times 100$$

$$= 10.1\%$$

Company B Ltd. :

$$\bar{X} = B + \frac{\sum d_Y}{n}$$

$$= 114 + \frac{0}{8} = 114$$

$$\sigma_Y = \sqrt{\frac{\sum d_Y^2}{n} - \left(\frac{\sum d_Y}{n}\right)^2}$$

$$= \sqrt{\frac{16}{8}} = \sqrt{2} = 1.414$$

$$C.V.\,(Y) = \frac{\sigma_Y}{\bar{Y}} \times 100$$

$$= \frac{1.414}{114} \times 100$$

$$= 1.24\%$$

$\because$ *C.V. (X)* is more than *C.V.(Y)*, the share prices of *A* are more variable.

Example 41. The runs scored by two batsmen *A* and *B* in 9 consecutive matches are given below :

A	:	85	20	62	28	74	5	69	4	13
B	:	72	4	15	30	59	15	49	27	36

Which of the batsmen is more consistent? **[*Delhi Univ. B.Com. (P) 2012*]**

Solution.

A			B		
X	$X - \bar{X}$	$(X - \bar{X})^2$	Y	$Y - \bar{Y}$	$(Y - \bar{Y})^2$
85	45	2025	72	39	1521
20	− 20	400	04	− 29	841
62	22	484	15	− 18	324
28	− 12	144	30	− 3	9
74	34	1156	59	26	676
05	− 35	1225	15	− 18	324
69	29	841	49	16	256
04	− 36	1296	27	− 6	36
13	− 27	729	26	− 7	49
$\sum X = 360$		$\sum (X - \bar{X})^2 = 8300$	$\sum Y = 297$		$\sum (Y - \bar{Y})^2 = 4036$

$$\bar{X} = \frac{\sum X}{n} = \frac{360}{9} = 40$$

$$\sigma_X = \sqrt{\frac{\sum (X - \bar{X})^2}{n}}$$

$$= \sqrt{\frac{8300}{9}} = \sqrt{922.22} = 30.37$$

$$C.V.\ (A) = \frac{\sigma_X}{\bar{X}} \times 100$$

$$= \frac{30.37}{40} \times 100 = 75.92\%$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{297}{9} = 33$$

$$\sigma_Y = \sqrt{\frac{\sum (Y - \bar{Y})^2}{n}}$$

$$= \sqrt{\frac{4036}{9}} = \sqrt{448.44} = 21.18$$

$$C.V.\ (B) = \frac{\sigma_Y}{\bar{Y}} \times 100$$

$$= \frac{21.18}{33} \times 100 = 64.18\%$$

Since $C.V.\ (B)$ is less than $C.V.\ (A)$, Batsman B is considered to be more consistent.

Example 42. For two Firms A and B, the following details are available :

	A	B
Number of workers	100	200
Average Salary of workers (₹)	1600	1750
Variance of salary of workers	256	289

Compute the following:

(i) Which firm pays larger pay package of salary to workers?

(ii) Which firm shows greater variability in the distribution of salary to the workers?

(iii) Compute the combined average salary and combined variance of all the 300 workers of both the firms taken together. **[*Delhi Univ. B.Com. (P) 2012*]**

Solution. Let n_1, n_2 denote the number of workers; $\bar{X}_1$, $\bar{X}_2$ denote the average salaries and σ_1, σ_2, the standard deviations of salaries of workers in Firm A and Firm B respectively. Then we are given:

$n_1 = 100 \quad \bar{X}_1 = 1600 \quad \sigma_1^2 = 256 \Rightarrow \sigma_1 = 16$

$n_2 = 200 \quad \bar{X}_2 = 1750 \quad \sigma_2^2 = 18 \Rightarrow \sigma_2 = 17$

(*i*) Total salary paid by Firm $A = n_1\bar{X}_1 = 100 \times 1600 = ₹1,60,000$

Total salary paid by Firm $B = n_2\bar{X}_2 = 200 \times 1750 = ₹ 3,50,000$

Thus the Firm *B* pays out larger package of salary.

(*ii*) To compare variability, we compute coefficients of variation:

$$C.V.\text{ (Firm } A) = \frac{\sigma_1}{\bar{X}_1} \times 100 = \frac{16}{1600} \times 100 = 1\%$$

$$C.V.\text{ (Firm } B) = \frac{\sigma_2}{\bar{X}_2} \times 100 = \frac{17}{1750} \times 100 = 0.97\%$$

Since *C.V.* (Firm *B*) is less than *C.V.* (Firm *A*), Firm *A* shows a greater variability in the distribution of salary.

(*iii*) Let $\bar{X}$ denote the combined average salary of all the employees in both the firms. Then $\bar{X}$ is given by

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2} = \frac{100 \times 1600 + 200 \times 1750}{100 + 200} = \frac{160000 + 350000}{300}$$

$$= \frac{510000}{300} = ₹1700$$

The combined variance, σ^2, of both firms is given by

$$\sigma^2 = \frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2},$$

where $d_1 = \bar{X}_1 - \bar{X} = 1600 - 1700 = -100 \Rightarrow d_1^2 = (-100)^2 = 10000$

and $d_2 = \bar{X}_2 - \bar{X} = 1750 - 1700 = 50 \Rightarrow d_2^2 = (50)^2 = 2500$

$$\therefore \quad \sigma^2 = \frac{100\,(256 + 10000) + 200\,(289 + 2500)}{100 + 200} = \frac{1025600 + 557800}{300}$$

$$= \frac{1583400}{300} = ₹5278.$$

Example 43. An analysis of the monthly salaries paid to employees in two companies *X* and *Y*, belonging to the same industry, provides the following results:

	Company X	*Company Y*
Number of employees	1200	1500
Average Salary (₹)	12000	9000
Standard deviation of salary (₹)	200	225

Find out the following from the above data :

(*i*) Combined average monthly salary and the combined standard deviation of salaries of the two companies.

(*ii*) The company having greater variability in individual salaries.

[*Delhi Univ. B.Com. (H) 2001*]

Solution. Let n_1, n_2 denote the number of employees; $\bar{X}_1$, $\bar{X}_2$ denote the average salaries and σ_1, σ_2 the standard deviations of salaries of employees in Company X and Company Y respectively. Then we are given :

$n_1 = 1200$ $\quad \bar{X}_1 = 12000$ $\quad \sigma_1 = 200$

$n_2 = 1500$ $\quad \bar{X}_2 = 9000$ $\quad \sigma_2 = 225$

(i) Let $\bar{X}$ denote the combined average salary of all the employees in both the companies. Then $\bar{X}$ is given by :

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2} = \frac{1200 \times 12000 + 1500 \times 9000}{1200 + 1500}$$

$$= \frac{14400000 + 13500000}{2700} = \frac{27900000}{2700} = ₹10333 \text{ (app)}$$

The combined standard deviation, σ, of both the companies is given by

$$\sigma = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}},$$

where $d_1 = \bar{X}_1 - \bar{X} = 12000 - 10333 = 1667$

and $d_2 = \bar{X}_2 - \bar{X} = 9000 - 10333 = -1333$

$$\therefore \quad \sigma = \sqrt{\frac{1200\,(40000 + 2778889) + 1500\,(50625 + 1776889)}{1200 + 1500}}$$

$$= \sqrt{\frac{3382666800 + 2741271000}{2700}} = \sqrt{\frac{6123937800}{2700}} = \sqrt{2268125.11}$$

$= ₹1506.03$

(ii) To compare variability, we compute coefficient of variation :

$$C.V. \text{ (Company } X) = \frac{\sigma_1}{\bar{X}_1} \times 100 = \frac{200}{1200} \times 100 = 1.67\ \%$$

$$C.V. \text{ (Company } Y) = \frac{\sigma_2}{\bar{X}_2} \times 100 = \frac{225}{9000} \times 100 = 2.5\ \%$$

Since $C.V.$ (Company X) is less than $C.V.$ (Company Y), the company Y shows greater variability in the distribution of salary.

Example 44. (i) Coefficients of variation of two series are 60% and 80%. Their standard deviations are 24 and 20 respectively. What are their arithmetic means?

[Delhi Univ. B.Com. (P) 1997]

(ii) Comment on the truthfulness of the following data :

For a set of 10 observations :

Mean = 5, $\quad S.D. = 2$ $\quad$ and $\quad C.V. = 60\%$

[Delhi Univ. B.Com. (P) 2003]